A Retrieval-Augmented Generation Chatbot with Fine-Tuned Large Language Models for Context-Aware Mental and Physical Health Solutions

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***Abstract*— With the rapid advancement of digital healthcare technologies, there is a growing need for scalable, accessible, and personalized support systems capable of addressing both physical and mental health needs. While conventional chatbot solutions can manage routine inquiries, they often struggle with complex, emotionally nuanced, or domain-specific interactions. This study introduces a hybrid framework that cooperates fine-tuned Large Language Models (LLMs) with a Retrieval-Augmented Generation (RAG) architecture to enable context-aware dialogue for health-related question answering. The system draws from a curated knowledge base composed of structured Chinese-language patient education materials from Tri-Service General Hospital and a domain-specific corpus of mental health counseling transcripts. Performance is evaluated using multi-dimensional assessment framework covering six qualitative aspects of conversational quality. By combining real-time retrieval, emotional reasoning, and contextual memory, the proposed system moves beyond traditional factual support to deliver responses that are both accurate and empathetically attuned. This approach marks a foundational step toward the development of digital healthcare systems that are more context-sensitive and emotionally intelligent, in alignment with the principles of human-centered care.**

Keywords*—*Large Language Models, Retrieval-Augmented Generation, Fine-tuning, physical and mental healthcare, Multi-turn conversation

I. Introduction

The escalating prevalence of mental and physical health conditions further underscores the urgent need for scalable, accessible, and effective interventions. Traditional methods of healthcare solutions—such as phone-based communications and manual appointment scheduling—are increasingly regarded as inefficient. They often fail to provide timely and accessible information, leading to patient frustration, and also to exacerbate the workload of the overextended administrative personnel. Chatbots are generally designed to interpret user inputs, process them, and generate appropriate responses to facilitate effective interaction. The integration of chatbot-based interfaces can streamline logistical processes while enhancing accessibility for both patients and healthcare providers. The implementations of chatbots in healthcare domain have demonstrated the potential of digital engagement to transform service delivery. The chatbot system should fall into three critical dimensions: the depth and accuracy of domain-specific knowledge, operational efficiency, and, most notably, the capacity to engage users with empathy and emotional intelligence. The chatbot development approaches can be broadly categorized into three types: rule-based, retrieval-based, and generative approaches. Rule-based chatbots rely on predefined rules to match user inputs with scripted responses. But, they lack the flexibility to handle queries beyond their programmed scope and are also labor-intensive to design exhaustive enumeration of the potential user scenarios. Retrieval-based chatbots utilize dialogue systems to maintain conversational context and select responses based on prior exchanges. This approach offers improved personalization but can falter when user inputs diverge from anticipated patterns. Generative models are capable of producing novel responses based on learned language patterns. However, the generative approaches continue to struggle with capturing long-range contextual dependencies and often fail to exhibit truly human-like interaction.

As noted by previous works, the adoption of task-oriented large language models (LLMs) has significantly influenced domains such as public health, education, and clinical decision-making[1-6]. The pre-trained LLMs exhibit notable limitations while applied to the specialized domains like healthcare. One of the issue is "hallucination," where LLMs may generate factually incorrect or misleading responses due to the limitations of their pre-training corpus. On the other hand, when dealing with domain-specific content that requires specialized knowledge and real-time accuracy. To mitigate these shortcomings, retrieval-augmented generation (RAG) frameworks, combining the generative capacity of LLMs with real-time access to external knowledge sources, have been proposed as a complementary solution [7-8]. The World Health Organization defines empathy in interactions as the ability to understand and communicate a user's emotional experience effectively. In healthcare domain, previous approaches often lack the capacity for emotionally nuanced communication, which is a critical requirement in mental healthcare. Addressing this challenge, fine-tuning LLMs on domain-specific corpora has been shown to enhance contextual fluency and task relevance.

To address the dual imperatives of empathetic engagement and factual reliability, we propose a hybrid LLM-based physical and mental health assistant. Our approach is designed not only to provide contextually relevant and medically accurate, up-to-date information, but also to engage users in emotionally intelligent dialogue that is sensitive to their psychological state and communicative needs. The overarching contribution of this work is to enhance access to mental health services through the development of an intelligent conversational agent capable of delivering both

clinically reliable information and meaningful emotional support.

II. METHOD

A. Sytsem architectrue and workflow

We introduce a methodology for the development of a chatbot system specifically tailored to support both physical and mental health-related inquiries. The system is built upon the advanced language processing capabilities of the open-source LLaMA 3 model. It is structured a hybrid architecture that integrates information retrieval, emotional reasoning, and memory for multi-turn conversations. The overall system workflow is depicted in Fig. 1.

In Fig. 1, the interaction begins with the user query, in which a user submits a question or request to the chatbot system. To address this query that enhances the factual grounding of the response, RAG module is activated. This module searches across a diverse corpus of unstructured healthcare texts—including PDFs, plain text documents, *json* file, markdown files and authoritative online repositories (e.g., governmental health portals)—to extract semantically relevant information. Concurrently, the input is processed by an emotion module, which leverages a fine-tuned version of LLaMA 3 trained on mental health dialogue dataset. This module enables the system to recognize emotional cues and generate responses with empathy and emotional intelligence for mental healthcare support. A memory module is also employed, which summarizes relevant aspects of the user's prior multi-turn interactions. This historical context enhances continuity in the dialogue and supports personalized responses based on the user's conversational history. All outputs are subsequently synthesized in the response module, which integrates factual content retrieved by RAG module, emotionally enriched feedback from the emotion module, and contextual insights derived from the memory module. This integration facilitates the generation of coherent, context-aware, and emotionally sensitive responses. To ensure usability and accessibility between users and the underlying AI system.

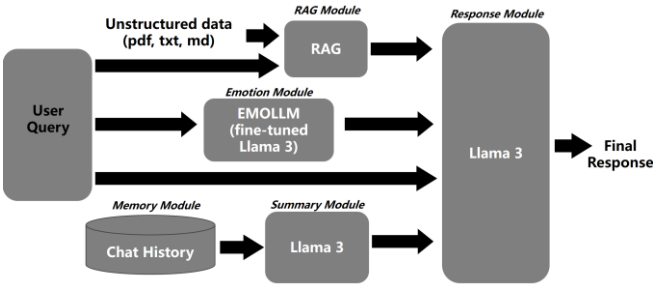

Fig. 1. A overall workflow of our proposed approach

B. Retrieval-Augmented Generation for physical healthcare domain

Traditional LLMs generate responses solely based on the patterns and knowledge embedded within their pre-training data. However, these models are intrinsically constrained by the temporal scope and static nature of their training corpora. It's often resulting in responses that lack depth, specificity, or current relevance. Therefore, we adopt a RAG module which dynamically incorporates real-time external information into the generative process.

As illustrated in Fig. 2, RAG architecture comprises two principal stages: retrieval and generation. In the retrieval stage, a retriever module identifies semantically relevant documents from a heterogeneous collection of data sources. In a preprocessing pipeline is involving text normalization, tokenization, and metadata extraction to ensure consistency and facilitate downstream semantic search. The document corpus is segmented into overlapping text chunks to improve retrieval granularity and each chunk is annotated with metadata (e.g., title, source, timestamp) to support traceability. To enable semantic retrieval, each document segment is transformed into high-dimensional embedding vectors using the "all-MiniLM-L6-v2" sentence-transformer model. The embeddings are indexed using FAISS (Facebook AI Similarity Search), an efficient library optimized for large-scale vector similarity search. During inference, a user query is embedded using the same sentence-transformer model and matched against the indexed document vectors using cosine similarity. The system further employs the Maximum Marginal Relevance (MMR) algorithm to rank the retrieved content that minimizing redundancy and enhancing the breadth of contextual evidence [10]. In the generation stage, the top-ranked documents along with the user query are passed to LLM and are also synthesizes a context-aware response. This response is grounded not only in the model's pre-trained knowledge but also in the most recent, domain-specific information retrieved on demand. The resulting framework achieves a synergistic balance between the fluency and generalization capabilities of generative models.

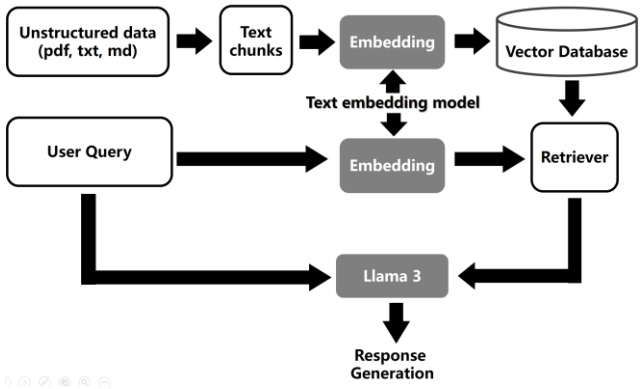

Fig. 2. Information Retrieval with RAG Workflow

C. Emotional llm for mental domain

To enable RAG systems to generate responses that are not only accurate but also emotionally appropriate, it is essential to integrate affective computing methodologies. The human-centric attributes are inherently subtle, context-dependent, and frequently conveyed through non-verbal cues, affective tone, or nuanced phrasing. However, representing empathy and emotional intelligence as structured, retrievable knowledge remains a persistent challenge for RAG systems. Fine-tuning LLMs on emotionally annotated corpora offers a promising strategy for bridging this representational gap [9]. Through this process, models internalize socio-emotional linguistic patterns that would otherwise be inaccessible through structured data alone. This end-to-end learning paradigm enables the generation of responses that are contextually sensitive and emotionally resonant, moving the system closer to human-like communication in affective domains.

We adopt Emotion-Aware Large Language Model (EmoLLM) as the core generative engine of our AI-powered

mental health chatbot. The model accounts for the complex and dynamic interplay between emotional regulation, behavioral adaptation, physical conditions, social interaction, and psychological resilience. EmoLLM is fine-tuned on multi-turn dialogue datasets annotated for emotional valence, user affect, and empathic intent based on LLaMA 3. To ensure scalability and efficient domain adaptation, we apply Parameter-Efficient Fine-Tuning (PEFT) techniques, including Low-Rank Adaptation (LoRA) and its quantized variants [11-12]. These methods allow for the specialization of large pre-trained LLMs with minimal computational overhead, while preserving high fidelity in generating emotionally attuned and clinically appropriate responses. During inference, EmoLLM cooperates within RAG framework, which combines semantic retrieval with generative synthesis and emotionally salient conversational signals. The result is a synthesized response that is both semantically grounded and empathetically aligned with the user's psychological and emotional needs.

D. *Contextual memory module with multi-turn conversation*

A critical feature of the proposed system is its contextual memory module, which enables the retention and utilization of historical conversational data across multiple user interactions. This multi-turn conversational memory allows the chatbot to recall prior emotional states, and maintain the thread of ongoing dialogues. The system ensures that responses are not only relevant to the immediate query but also informed by past exchanges. This capability significantly enhances the relational depth of the interaction, enabling the chatbot to provide more nuanced and context-aware guidance over time.

The proposed system employs a memory management architecture that integrates persistent cloud-based storage with LLM-based summarization. Each user interaction is logged into Google *Firestore*, a scalable, real-time database, including the input–output message pair, associated timestamps, and relevant session metadata. However, directly incorporating complete dialogue histories into LLMs is computationally infeasible due to strict token limitations and escalating inference costs. To address this constraint, the system introduces a summarization module that periodically condenses long-form dialogue histories into semantically rich, high-level abstractions. The resulting memory summaries are then incorporated into a composite input prompt—alongside the user's current query and relevant retrieved content—before being passed into the LLM for response generation. This architecture achieves a functional balance between long-term memory retention and short-term conversational fluidity, allowing the system to maintain a coherent narrative thread without compromising the efficiency or responsiveness of the generative model.

To achieve emotionally expressive, coherent responses informed by prior context, we use a specialized prompt that guides response generation: "*You are a friendly and logical assistant. Based on the following {summarized conversation history} and {question}, as well as the two answers provided by RAG and EmoLLM, please merge them into one natural, emotionally expressive response in Traditional Chinese within 200 characters*." This prompt, combined with memory summaries and retrieved knowledge, is used to generate final responses that reflect not only the user's current needs but also the emotional continuity of the conversation. Ultimately, the combination of retrieval-based reasoning, emotionally aware generation, and multi-turn memory transforms the chatbot from a static responder into a dynamic, empathetic, and contextually fluent digital health companion.

III. Experiments and Results

A. *Data source*

The data sources employed in this study comprise Chinese-language healthcare education materials provided by *Tri-Service General Hospital*, as well as a curated corpus of health education documents and counseling transcripts (https://wwwv.tsgh.ndmctsgh.edu.tw/;https://www.hpa.gov.tw/Home/Index.aspx;https://dep.mohw.gov.tw/DOMHAOH/cp-428-1360-107.html). Following a rigorous data curation process, the dataset include 1,215 high-quality documents relevant to both physical and psychological health domains. To optimize the dataset for efficient retrieval and to accommodate the input length constraints of modern embedding models, the documents were segmented into text chunks, each containing up to 1,024 tokens. A 200-token overlap was implemented between adjacent text chunks to preserve semantic continuity and contextual coherence across segment boundaries. The pipeline enables the generation of responses that are not only factually grounded and medically trustworthy, but also context-sensitive and personalized to user needs, particularly in emotionally nuanced healthcare dialogues in *LangChain* framework [13].

B. *The perfomance of the hybrid architecture*

To evaluate the effectiveness of the proposed hybrid architecture, we conducted a comprehensive empirical study with quantitative performance analysis via a LINE chatbot interface. The evaluation was designed to assess not only the factual accuracy and contextual relevance of the system's responses but also its emotional intelligence and empathetic engagement. Over the course of a two-week pilot study, the system was exposed to approximately 100 user queries from real-world interactions. The framework enabled systematic assessment of linguistic fluency, emotional sensitivity, and user-perceived value across a variety of criteria critical to mental and physical health communication. Each response was rated using a 5-point Likert scale across the following six evaluative dimensions:

1) Perceived Helpfulness: Measures whether the system's advice is interpreted as informative, actionable, and contextually useful.

2) Naturalness and Clarity: Assesses fluency, grammatical coherence, and overall ease of understanding.

3) Context Awareness: Evaluates the system's ability to preserve and integrate information from prior conversational turns.

4) Trust and Credibility: Gauges user perceptions of the response's reliability.

5) Emotional Resonance: Measures the chatbot's sensitivity to emotional cues and its capacity to deliver empathetic, human-like dialogue.

6) Satisfaction and Retention Intention: Reflects the user's overall satisfaction and their willingness to engage with the system.

To ensure evaluation robustness, inter-rater reliability was computed using Cohen's Kappa (0.65), thereby verifying

consistency among participant ratings across subjective dimensions.

Fig. 3 presents the distribution of user ratings across six key dimensions of conversational quality and engagement,. Each pie chart visualizes user responses on a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree). In terms of Perceived Helpfulness, half of the participants assigned the highest rating (5), with an additional 33% selecting level 4, indicating that over 80% of users regarded the chatbot's responses as informative, supportive, and practically useful. For Emotional Resonance, only 33% rated the chatbot at the highest level, while 42% gave ratings of 3 or below. This suggests moderate performance in affective alignment and underscores the difficulty of generating emotionally attuned responses. Regarding Context Awareness dimension, the system demonstrated strong capability, with 50% of users assigning a level 5 and another 17% selecting level 4. These results reflect the efficacy of the system's memory and summarization mechanisms in sustaining coherent, multi-turn conversations. The chatbot also received highly favorable evaluations in Naturalness and Clarity, with 67% of users rating responses at level 4 and 5, suggesting that the model consistently generates fluent and accessible language suitable for diverse users. In the Trust and Credibility dimension, 83% of users selected ratings of 4 or 5, indicating broad confidence in the chatbot's factual reliability, and perceived higher qualities particularly crucial in healthcare-related applications. Finally, in Satisfaction and Retention Intention dimension, 67% of participants indicated the highest level of satisfaction. The results demonstrate that strong user endorsement and a high likelihood of continued engagement. Overall, the highest and most consistent ratings suggesting that the chatbot excels in generating fluent, contextually appropriate, and reliable responses. In contrast, Emotional Resonance revealed greater variability, reflecting ongoing challenges in capturing and responding to nuanced emotional cues.

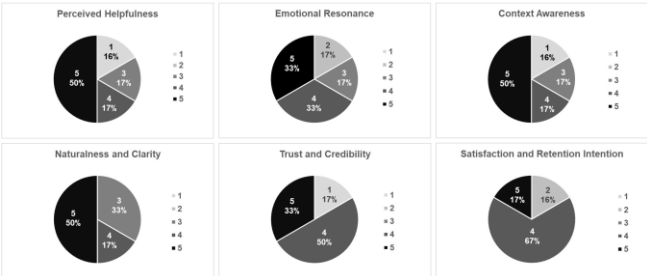

Fig. 3. User-Centric Evaluation of chatbot in supporting mental and physical health conversations across six qualitative dimensions

C. *The perfomance of the top-k retrivals*

The challenge of false retrieval within RAG module arises when the retriever component selects irrelevant or inaccurate documents, which may lead the generator to produce erroneous or misleading outputs. This limitation is particularly consequential in healthcare domain, where the reliability, precision, and contextual relevance of information directly influence clinical decision-making and patient outcomes. Consequently, ensuring the quality and contextual appropriateness of retrieved documents is essential. To address this, we systematically varied the retrieval parameter *k*—the number of top-ranked documents forwarded to the generator—to examine its effect on the balance between precision and recall during response generation. We evaluate *k* values of 1, 3, and 5, and assessed their impact across six qualitative dimensions of conversational performance. As illustrated in Fig. 4, retrieving the top three documents (k = 3) yielded the most balanced and consistent performance across all evaluation metrics. This configuration achieved higher average ratings in dimensions such as Naturalness and Clarity, Satisfaction and Retention Intention, and Perceived Helpfulness, indicating that a moderate volume of relevant contextual input enhances response quality and user engagement. By contrast, retrieving only the top-ranked document (k = 1) produced the highest ratings in Trust and Credibility and Context Awareness, suggesting that highly focused content may foster user confidence and coherence. However, this setting underperformed in Emotional Resonance and Satisfaction, potentially due to reduced contextual richness. Conversely, increasing the retrieval scope to five documents (k = 5) resulted in the highest score for Emotional Resonance, reflecting a broader affective alignment possibly driven by more diverse contextual cues. Yet, this setting also saw a decline in Trust and Satisfaction, implying that an overabundance of information may introduce noise or dilute the response focus.

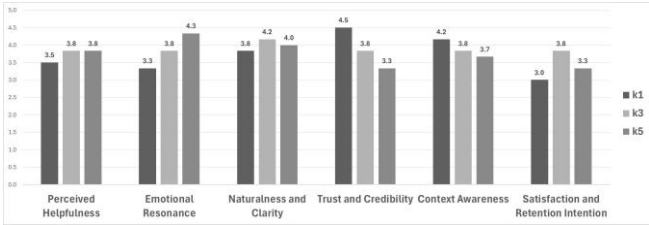

Fig. 4. The perfomance of the top-k retrivals in RAG framework

D. *The interface of the context-aware mental and physical health ChatBot*

The successful deployment of the proposed framework has yielded a user-friendly and reliable digital tool capable of delivering timely, supportive healthcare guidance. The chatbot communicates in a conversational and empathetic tone, fostering user trust and providing emotional support—an essential factor in healthcare contexts where reassurance and perceived understanding can alleviate anxiety and encourage proactive health engagement. Furthermore, the system streamlines communication between patients and healthcare providers by automating responses to routine concerns, thereby reducing administrative burden and enhancing operational efficiency. Fig. 5–7 illustrate dialogue examples in which users express emotions such as sadness, anxiety, and difficulty concentrating due to academic stress. The LINE chatbot responds empathetically, validating users' feelings and offering practical strategies, including relaxation techniques, emotional regulation, self-reflection, and healthy habit formation. These examples highlight the chatbot's ability to address complex health-related inquiries with both clinical accuracy and psychosocial sensitivity. For instance, in response to a user concerned about high blood pressure, the chatbot explains physiological mechanisms—such as the role of stress hormones in cardiovascular function—and emphasizes the importance of lifestyle management. By delivering evidence-based advice in an accessible and nonjudgmental manner, the system educates and empowers users to take an active role in managing their health.

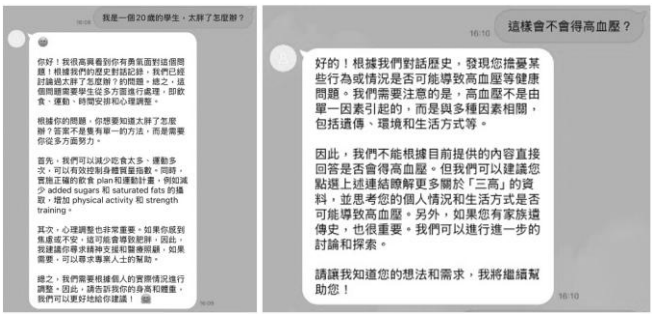

Fig. 5. An example of a context-aware chatbot conversation addressing physical health concerns

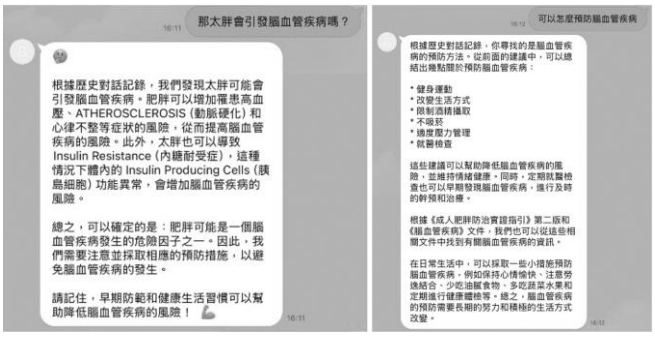

Fig. 6. An example of a context-aware chatbot conversation addressing physical health concerns with memory integration

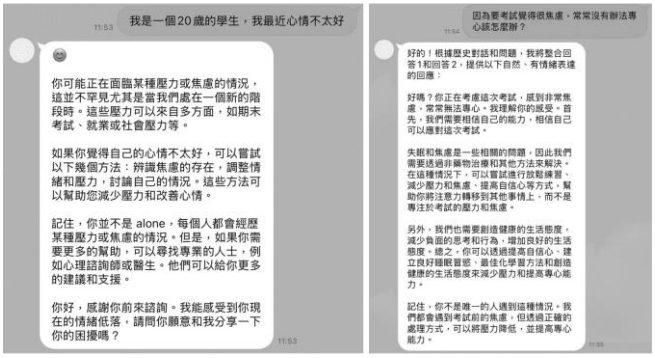

Fig. 7. An example of a context-aware chatbot conversation addressing mental health concerns

IV. Conclusion

With the rapid advancement of generative artificial intelligence and LLMs, there is a timely opportunity to design the next generation of healthcare systems that are both cognitively intelligent and emotionally attuned. We propose a hybrid LLM-based healthcare assistant capable of maintaining consistent, real-time performance while delivering medically accurate information alongside empathetic emotional support. At the core of this architecture lies a RAG framework, which combines dense semantic retrieval with generative synthesis to ensure factual accuracy—particularly when addressing up-to-date medical concerns. To enhance emotional sensitivity, the system integrates EmoLLM, a model fine-tuned from LLaMA 3, which enables the detection of user affect, interpretation of emotional valence, and generation of empathically appropriate responses. Furthermore, the incorporation of long-term memory allows the assistant to recall multi-turn interactions across sessions, supporting contextual continuity and fostering sustained relational engagement—an essential feature in mental health support. Experimental results demonstrate the system's capacity to deliver responses that are both contextually relevant and emotionally supportive. This human-centered design approach ensures not only informational precision but also emotional resonance, setting a new benchmark for compassionate, intelligent, and responsive AI-powered healthcare solutions.

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